

KEVIN SCHWARZWALD

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Interdisciplinary climate researcher helping inform climate impacts projections and decision-making under uncertainty

EDUCATION

Columbia University

Doctor of Philosophy, Earth and Environmental Sciences

New York, NY

October 2024

Master of Philosophy, Earth and Environmental Sciences

October 2023

Master of Arts, Earth and Environmental Sciences

May 2021

Research Areas: climate variability, uncertainty in projections of climate and climate impacts, precipitation in East Africa, tropical convection, climate model evaluation, the use of climate projections in policy and adaptation efforts

Advisors: Lisa Goddard (†2022), Richard Seager; *Committee Members*: Mingfang Ting, Kate Marvel, Radley Horton

- *PhD thesis*: *The Present and Future of the Horn of Africa Rains*

Peking University, Yenching Academy (北京大学燕京学堂)

Beijing, China

Master of Law in China Studies (concentration in Politics and International Relations)

July 2017

Honors: Awards for Outstanding Academic Achievement and Outstanding Contributions to the Academy

- *Master's thesis*: *The Relationship between Fiscal Systems and Inefficient Urban Expansion in the PRC: An Empirical Look at Prefecture-Level Urbanization*

The University of Chicago

Chicago, IL

Bachelor of the Arts (A.B.) in Physics; Public Policy (concentration in Economics)

June 2015

Honors: Graduated with General Honors, Dean's List 2011-2013, 2015, Departmental Honors in Public Policy

- *Senior Thesis (Public Policy)*: *"The Wrong Folks Kept Winning": Protectionism and America's Trains*
 - Study Abroad: Paris, France
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ACADEMIC RESEARCH EXPERIENCE

Columbia Climate School

September 2024 - present

Climate School Postdoctoral Scientist

New York, NY

Mentors: Radley Horton and Gernot Wagner

- Current projects include: understanding and constraining the range of climate uncertainty in projections of climate impacts in data-sparse regions; the impact of internal variability on projections of labor productivity; understanding the suitability of Global Warming Levels for regional climate impacts studies

International Research Institute for Climate and Society (IRI),

September 2019 – August 2024

Lamont-Doherty Earth Observatory (LDEO)

Graduate Research Assistant

New York, NY

- Projects included: the present and future of rainfall dynamics in East Africa, understanding the sources of climate uncertainty in projections of the societal impacts (agriculture, mortality, etc.) of climate change, and building tutorials and python tools to make climate data more accessible to end users

Energy Policy Institute at the University of Chicago (EPIC)

September 2017 – July 2019

Pre-Doctoral Fellow (joint with RDCEP)

Chicago, IL and remotely from New York, NY as GISS Visiting Researcher

- Conducted interdisciplinary research on how climate variability will impact society (agricultural yields, mortality, energy consumption) and how to improve weather projections used in climate impacts studies, with Prof. Amir Jina
- Adapted latest downscaling and bias-correction methods to impacts studies, sensitivity tested common economic damage functions to climate variability changes, quantified effect of internal climate uncertainty on climate damages
- Advised on the use of climate model and observational data in economics and policy applications

Center for Robust Decision-Making on Climate and Energy Policy (RDCEP)

April 2015 – July 2019

Pre-Doctoral Fellow (joint with EPIC 2017-2019)

Chicago, IL and remotely from New York, NY, and Beijing, China

- Researched changes in precipitation and temperature variability from synoptic to inter-decadal timescales in global climate models in Prof. Elisabeth Moyer's climate statistics group
 - Co-organized a weekly interdisciplinary speaker series on energy and environmental themes attended by ~25 young researchers and students to foster collaboration across geophysics and economics research communities
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PUBLICATIONS

Schwarzwald, Kevin and Richard Seager. 2024. "Revisiting the 'East African Paradox': CMIP6 models also struggle to reproduce strong observed MAM drying trends." *Journal of Climate* (Early Online Release). [Link](#).

- Schwarzwald, Kevin**, Richard Seager, Mingfang Ting, and Alessandra Giannini. 2023. “Large-Scale Stability and the Greater Horn of Africa Long and Short Rains.” *Journal of Climate* 36 (20): 7297–7317. [Link](#).
- Anderson, Weston, Benjamin I. Cook, Kim Sliniski, **Kevin Schwarzwald**, Amy McNally, and Chris Funk. 2023. “Multiyear La Niña Events and Multiseason Drought in the Horn of Africa.” *Journal of Hydrometeorology* 24 (1): 119–31. [Link](#).
- Schwarzwald, Kevin**, and Nathan Lenssen. 2022. “The Importance of Internal Climate Variability in Climate Impact Projections.” *Proceedings of the National Academy of Sciences* 119 (42). [Link](#).
- Schwarzwald, Kevin**, Lisa Goddard, Richard Seager, Mingfang Ting, and Kate Marvel. 2022. “Understanding CMIP6 Biases in the Representation of the Greater Horn of Africa Long and Short Rains.” *Climate Dynamics* 61 (3): 1229–55. [Link](#).
- Schwarzwald, Kevin**, Andrew Poppick, Maria Rugenstein, Jonah Bloch-Johnson, Jiali Wang, David McInerney, and Elisabeth J. Moyer. 2021. “Changes in Future Precipitation Mean and Variability across Scales.” *Journal of Climate* 34 (7): 2741–58. [Link](#).
- Schwarzwald, Kevin** and Kerrie Geil. “xagg: A Python package to aggregate gridded data onto polygons.” *Under review at the Journal of Open Source Software*. [Link](#).

AWARDS, GRANTS, COMPETITIVE PROGRAMS

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| - CATER School on Transdisciplinary Climate Risk Adaptation and Action (Naivasha, Kenya; fully funded, including travel) | November 2023 |
| - ASciNA Young Scientist Award, \$7500 | 2023 |
| - Outstanding Oral Presentation Award, 103 rd AMS Annual Meeting | 2023 |
| - Center for Science and Society Seed Grants, \$3975 total (to NENSIC, co-PI) | 2021, 2022, 2023 |
| - ICTP Summer School on Theory, Mechanisms and Hierarchical Modeling of Climate Dynamics (Trieste, Italy; fully funded) | 2022 |
| - Yenching Academy Dean’s Research Grant, ¥5000 | 2016 |
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OTHER EXPERIENCE

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| Eparque Urban Strategies
<i>Independent Consultant (part-time)</i> | March 2019 – July 2019
New York, NY |
| Yenching Global Symposium (全球青年中国论坛)
<i>Associate Director, Logistics</i> | October 2015 – April 2016
Beijing, China |
| Technology Policy Institute
<i>Google Policy Fellow</i> | July 2014 – August 2014
Washington, DC |
| Paris Center for Cosmological Physics
<i>Lab Intern</i> | January 2014 – March 2014
Paris, France |
| alpS GmbH
<i>Management Intern, Consulting Group</i> | August 2013 – September 2013
Innsbruck, Austria |
| Moser/Jaritz
<i>Research Intern</i> | August 2013 – September 2013
Gmunden, Austria |
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INVITED AND CONTRIBUTED PRESENTATIONS

- 2024:** WCRP EPESC Science Meeting, JPL Science Understanding through Data Science (SUDS) Seminar, European Geosciences Union (EGU)’s General Assembly, Max Planck Institute for Meteorology Hamburg, Oxford School of Geography and the Environment
- 2023:** American Meteorological Association (AMS)’s Annual Meeting; EGU General Assembly; University of California Santa Barbara (UCSB) Climate Hazards Center; ECS Symposium; WCRP Open Science Conference; American Geophysical Union (AGU) Fall Meeting (**invited talk**)
- 2022:** AGU Fall Meeting; Institute for Mathematical and Statistical Innovation (IMSI); NASA Goddard Institute for Space Studies (GISS); ICTP Summer School on Theory, Mechanisms and Hierarchical Modeling of Climate Dynamics; Interdisciplinary PhD Student Workshop in Sustainable Development (IPWSD); Columbia Student Climate Symposium
- 2021:** AGU Fall Meeting; University of Chicago Environmental Data Science Lunch
- 2019:** US CLIVAR Large Ensembles Workshop
- 2018:** AGU Fall Meeting

DATA TOOLS AND TUTORIALS

Weather Panel Tutorial

climateestimate.net

- In an inter-disciplinary team, co-developed a tutorial designed to introduce climate economists to working with observational climate data using python (xarray), R, or MATLAB

xagg

github.com/ks905383/xagg

- Lead developer and maintainer of **xagg**, a python package for aggregating raster data (such as gridded weather data) onto polygons (such as administrative shapefiles), using area-averaging based on the overlap between pixels, polygons

quantproj

github.com/ks905383/quantproj

- Developer of **quantproj**, an R package for projecting weather data by using downscaled fine-scaled distributional changes in Large Ensembles and the delta method; based on the methodology of Haugen et al., 2018
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SKILLS

Computer (Analysis): advanced proficiency in python (xarray, dask), MATLAB, R, and Excel; basic proficiency in GIS (QGIS) and Stata; experience with high-performance computing (code optimization, parallel processing, batch scheduling, etc.), bash scripting; experience with climate (NetCDF, nco, CMIP, etc.) and satellite data analysis (DMSP/OLS, VIIRS, etc.)

Computer (Modeling): experience in running SPEEDY, an intermediate-complexity global climate model (GCM)

Computer (Communication/Design): advanced proficiency in LaTeX, PowerPoint, Keynote, inkscape; basic proficiency in Adobe Illustrator

Language: native in German, professionally fluent in French, basic communication skills in Mandarin Chinese

PROFESSIONAL NETWORKS

- American Geophysical Union (AGU)
- European Geosciences Union (EGU)
- Austrian Scientists & Scholars in North America (ASciNA)